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$$f(x, y) = -3x^2 - 2y^2 + 12x - 12y + 10$$

$$\begin{cases} \frac{\partial f}{\partial x} = -6x + 12 = 0 \\ \frac{\partial f}{\partial y} = -4y - 12 = 0 \end{cases} \Rightarrow \begin{cases} -6(x - 2) = 0 \\ -4(y + 3) = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = -3 \end{cases}$$
$$\Rightarrow (x, y) \in \{(2, -3)\}$$

$$\begin{cases} \frac{\partial^2 f}{\partial x^2} = -6 \\ \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 0 \\ \frac{\partial^2 f}{\partial y^2} = -4 \end{cases} \Rightarrow H(x, y) = 24$$

- $H(2, -3) = 24 > 0$ en $\frac{\partial^2 f}{\partial x^2}(2, -3) = -6 < 0 \Rightarrow (2, -3)$ maximum

References